

$\int_a^b f(x)dx$
 $\lim_{x \rightarrow 0} \frac{f(x)}{x}$
 $\frac{dy}{dx}$
 $ax+by \leq c$
 $\sqrt{x^2+y^2}$

UNIT 3

INTEGRATION

Introduction:

The technique or method to find such a function whose derivative is given involves the inverse process of differentiation, called anti-derivation or integration.

Differentials of Variables:

Let f be a differentiable function in the interval $a < x < b$, defined as $y = f(x)$, then

the differential of independent variable x is denoted by dx and is defined to be the increment δx

Thus $\delta x = dx$

The differential of dependent variable y is denoted by dy and is written as

$$dy = f'(x)dx \text{ and } \delta y \approx dy$$

EXERCISE 3.1

Q.1 Find δy and dy in the following cases

(i) $y = x^2 - 1$, when x changes from 3 to 3.02

Solution:

$$x = 3 \text{ and } dx = \delta x = 3.02 - 3 = 0.02$$

$$y = x^2 - 1$$

$$y + \delta y = (x + \delta x)^2 - 1$$

$$\delta y = (x + \delta x)^2 - 1 - (x^2 - 1)$$

$$= x^2 + (\delta x)^2 + 2x\delta x - 1 - x^2 + 1$$

$$= \delta x(\delta x + 2x)$$

$$= (0.02)(0.02 + 2(3))$$

$$= (0.02)(6.02)$$

$$\boxed{\delta y = 0.1204}$$

$$y = f(x) = x^2 - 1$$

$$f'(x) = 2x$$

$$dy = f'(x)dx$$

$$dy = 2x dx \quad \because f'(x) = 2x$$

$$dy = 2(3)(0.02)$$

$$dy = (6)(0.02)$$

$$\boxed{dy = 0.12}$$

(ii) $y = x^2 + 2x$, when x changes from 2 to 1.8

Solution:

$$x = 2 \text{ and } dx = \delta x = 1.8 - 2 = -0.2$$

$$y = x^2 + 2x$$

$$y + \delta y = (x + \delta x)^2 + 2(x + \delta x)$$

$$\delta y = (x + \delta x)^2 + 2(x + \delta x) - (x^2 + 2x)$$

$$= x^2 + (\delta x)^2 + 2x\delta x + 2x + 2\delta x - x^2 - 2x$$

$$= (\delta x)^2 + 2x\delta x + 2\delta x$$

$$= (-0.2)^2 + 2(2)(-0.2) + 2(-0.2)$$

$$\delta y = 0.04 - 0.8 - 0.4$$

$$\boxed{\delta y = -1.16}$$

$$y = f(x) = x^2 + 2x$$

$$f'(x) = 2x + 2$$

$$dy = f'(x)dx$$

$$dy = (2x + 2)dx$$

$$dy = [2(2) + 2](-0.2)$$

$$dy = (6)(-0.2)$$

$$\boxed{dy = -1.2}$$

- (iii) $y = \sqrt{x}$ when x changes from 4 to 4.1

Solution:

$$x = 4 \text{ and } \delta x = dx = 4.41 - 4 = 0.41$$

$$y = \sqrt{x}$$

$$\text{Then } y + \delta y = \sqrt{x + \delta x}$$

$$\delta y = (x + \delta x)^{\frac{1}{2}} - x^{\frac{1}{2}}$$

$$\delta y = (4 + 0.4)^{\frac{1}{2}} - (4)^{\frac{1}{2}}$$

$$\delta y = \sqrt{4.41} - \sqrt{4}$$

$$\boxed{\delta y = 0.1}$$

$$y = f(x) = \sqrt{x}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$dy = f'(x)dx$$

$$dy = \frac{1}{2\sqrt{x}}.dx$$

$$dy = \frac{1}{2\sqrt{4}}(0.41)$$

$$dy = \frac{0.41}{4}$$

$$\boxed{dy = 0.1025}$$

Q.2 Using differentials, find $\frac{dy}{dx}$ and $\frac{dx}{dy}$

in the following equations

(i) $xy + x = 4$

Solution:

$$xy + x = 4$$

Taking differentials on both sides

$$d(xy + x) = d(4)$$

$$d(xy) + d(x) = 0$$

$$x dy + y dx + dx = 0$$

$$x dy = -(y + 1) dx \dots (i)$$

$$dy = -\left[\frac{y+1}{x}\right] dx$$

$$\boxed{\frac{dy}{dx} = -\frac{y+1}{x}}$$

From (i)

$$(y + 1) dx = -x dy$$

$$\boxed{\frac{dx}{dy} = -\frac{x}{y+1}}$$

(ii) $x^2 + 2y^2 = 16$

Solution:

$$x^2 + 2y^2 = 16$$

Taking differentials on both sides

$$d(x^2 + 2y^2) = d(16)$$

$$2x dx + 2(2y dy) = 0$$

$$2x dx + 4y dy = 0$$

$$2y dy = -x dx \dots (i)$$

$$dy = \left(\frac{-x}{2y}\right) dx$$

$$\boxed{\frac{dy}{dx} = \frac{-x}{2y}}$$

From (i)

$$x dx = -2y dy$$

$$dx = \left(\frac{-2y}{x}\right) dy$$

$$\boxed{\frac{dx}{dy} = \frac{-2y}{x}}$$

(iii) $x^4 + y^2 = xy^2$

Solution:

$$x^4 + y^2 = xy^2$$

Taking differentials on both sides

$$d(x^4 + y^2) = d(xy^2)$$

$$4x^3 dx + 2y dy = y^2 dx + x(2y dy)$$

$$2y dy - 2xy dy = y^2 dx - 4x^3 dx$$

$$2y(1-x) dy = (y^2 - 4x^3) dx \dots (i)$$

$$dy = \left[\frac{y^2 - 4x^3}{2y(1-x)} \right] dx$$

$$\boxed{\frac{dy}{dx} = \frac{y^2 - 4x^3}{2y(1-x)}}$$

From (i)

$$dx = \left(\frac{2y(1-x)}{y^2 - 4x^3} \right) dy$$

$$\boxed{\frac{dx}{dy} = \frac{2y(1-x)}{y^2 - 4x^3}}$$

(iv) $xy - \ln x = c$

Solution:

$$xy - \ln x = c$$

Taking differentials on both sides

$$d(xy - \ln x) = d(c)$$

$$d(xy) - d(\ln x) = 0$$

$$x dy + y dx - \frac{1}{x} dx = 0$$

$$x dy = \frac{1}{x} dx - y dx$$

$$x dy = \left(\frac{1}{x} - y \right) dx$$

$$dy = \frac{1}{x} \left(\frac{1-xy}{x} \right) dx$$

$$dy = \left(\frac{1-xy}{x^2} \right) dx \dots (i)$$

$$\boxed{\frac{dy}{dx} = \frac{1-xy}{x^2}}$$

From (i),

$$\frac{1}{x} dx - y dx = x dy$$

$$dx \left(\frac{1-xy}{x} \right) = x dy$$

$$\boxed{\frac{dx}{dy} = \frac{x^2}{1-xy}}$$

Q.3 Use differentials to approximate the values of

(i) $\sqrt[4]{17}$

Solution:

$$\text{Let } f(x) = \sqrt[4]{x} = (x)^{\frac{1}{4}}$$

$$\text{Then } f(x + \delta x) = (x + \delta x)^{\frac{1}{4}}$$

$$f(x + \delta x) = (x + dx)^{\frac{1}{4}} (\because \delta x = dx)$$

As the nearest perfect fourth root to 17 is 16 so, we take $x = 16$ and

$$\delta x = dx = 1$$

$$f(16) = (16)^{\frac{1}{4}} = 2$$

$$\text{As } f(x) = x^{\frac{1}{4}}$$

Differentiate w.r.t x

$$f'(x) = \frac{1}{4} x^{\frac{1}{4}-1} = \frac{1}{4} x^{-\frac{3}{4}}$$

$$f'(x) = \frac{1}{4x^{\frac{3}{4}}}$$

put $x = 16$ in

$$\begin{aligned} f'(16) &= \frac{1}{4(16)^{\frac{3}{4}}} = \frac{1}{4 \left[(16)^{\frac{1}{4}} \right]^3} \\ &= \frac{1}{4(2)^3} \end{aligned}$$

$$f'(16) = \frac{1}{32}$$

By using

$$f(x + \delta x) \approx f(x) + f'(x) dx$$

$$f(16+1) \approx f(16) + f'(16)(1)$$

$$f(17) \approx 2 + \frac{1}{32}$$

$$\sqrt[4]{17} \approx 2 + 0.03125$$

$$\boxed{\sqrt[4]{17} \approx 2.03125}$$

(ii) $(31)^{\frac{1}{5}}$

Solution:

Let $f(x) = x^{\frac{1}{5}}$

As the nearest perfect fifth root to 31 is 32, so we take $x = 32$ and $\delta x = dx = -1$

$$f(32) = (32)^{\frac{1}{5}} = 2$$

$$f(x) = x^{\frac{1}{5}}$$

Differentiate w.r.t x

$$f'(x) = \frac{1}{5} x^{\frac{1}{5}-1} = \frac{1}{5} x^{-\frac{4}{5}}$$

$$f'(x) = \frac{1}{5x^{\frac{4}{5}}}$$

Put $x = 32$ in

$$f'(32) = \frac{1}{5(32)^{\frac{4}{5}}} = \frac{1}{5 \left[(32)^{\frac{1}{5}} \right]^4}$$

$$= \frac{1}{5(2)^4}$$

$$f'(32) = \frac{1}{80}$$

By using

$$f(x + \delta x) \approx f(x) + f'(x)dx$$

$$f(32 + (-1)) \approx f(32) + f'(32)(-1)$$

$$f(31) \approx 2 - \frac{1}{80}$$

$$\sqrt[5]{31} \approx 2 - 0.0125$$

$$\boxed{\sqrt[5]{31} \approx 1.9875}$$

(iii) $\cos 29^\circ$

Solution:

Let $f(x) = \cos x$

We take

$$x = 30^\circ \text{ and}$$

$$\delta x = dx = -1^\circ = \frac{-\pi}{180} = -0.01745 \text{ radians}$$

$$f(30^\circ) = \cos 30^\circ = 0.8660$$

$$f(x) = \cos x$$

Differentiate w.r.t x

$$f'(x) = -\sin x$$

$$f'(30^\circ) = -\sin 30^\circ = -\frac{1}{2}$$

$$f(x + \delta x) \approx f(x) + f'(x)dx$$

$$f(30^\circ - 1^\circ) \approx f(30^\circ) + f'(30^\circ)(-0.01745)$$

$$f(29^\circ) \approx 0.8660 + \left(-\frac{1}{2}\right)(-0.01745)$$

$$\cos 29^\circ \approx 0.8660 + 0.0087$$

$$\boxed{\cos 29^\circ = 0.8747}$$

(iv) $\sin 61^\circ$

Solution:

Let $f(x) = \sin x$

We take

$$x = 60^\circ \text{ and}$$

$$\delta x = dx = 1^\circ = \frac{\pi}{180} = 0.01745 \text{ radians}$$

$$f(60^\circ) = \sin 60^\circ = 0.8660$$

$$f(x) = \sin x$$

Differentiate w.r.t x

$$f'(x) = \cos x$$

$$f'(60^\circ) = \cos 60^\circ = \frac{1}{2}$$

Using

$$f(x + \delta x) \approx f(x) + f'(x)dx$$

$$f(60^\circ + 1^\circ) \approx f(60^\circ) + f'(60^\circ)(0.01745)$$

$$f(61^\circ) \approx (0.8660) + \frac{1}{2}(0.01745)$$

$$\sin 61^\circ \approx 0.8660 + (0.5)(0.01745)$$

$$\boxed{\sin 61^\circ \approx 0.8747}$$

- Q.4 Find the approximate increase in the volume of a cube if the length of its each edge changes from 5 to 5.02**

Solution:

Let x be the length of each edge of cube and V be the volume of cube, then

$$V = x^3 \dots (i)$$

Here

$$x = 5, dx = 5.02 - 5 = 0.02$$

Taking differentials on both sides of (i)

$$dV = 3x^2 dx$$

$$dV = 3(5)^2 (0.02)$$

$$dV = 75(0.02)$$

$$\boxed{dV = 1.5}$$

Thus the approximate increase in volume of the cube is 1.5 cubic units

- Q.5 Find the approximate increase in the area of circular disc if its diameter is increased from 44 cm to 44.4 cm**

Solution:

Let r be the radius of the circle

then r changes from 22 cm to 22.2 cm

$$\text{So } r = 22, dr = 22.2 - 22 = 0.2$$

$$A = \pi r^2$$

Taking differentials of both sides

$$dA = \pi(2rdr)$$

$$dA = 2\pi(22)(0.2)$$

$$dA = 8.8\pi$$

$$\boxed{dA = 27.646}$$

Thus the approximate increase in area of circular disc is 27.646 cm^2